

```
[1]: numbers = [1, 2, 3, 4, 5]
squares = list(map(lambda x: x * x, numbers))
print("Original List:", numbers)
print("Squares List:", squares)
```

```
Original List: [1, 2, 3, 4, 5]
Squares List: [1, 4, 9, 16, 25]
```

```
[2]: numbers = [10, 15, 20, 25, 30, 35, 40]
even_numbers = list(filter(lambda x: x % 2 == 0, numbers))
print("Original List:", numbers)
print("Even Numbers:", even_numbers)
```

```
Original List: [10, 15, 20, 25, 30, 35, 40]
Even Numbers: [10, 20, 30, 40]
```

```
[4]: from functools import reduce
numbers = [1, 2, 3, 4, 5]
sum_result = reduce(lambda x, y: x + y, numbers)
print("List Elements:", numbers)
print("Sum of Elements:", sum_result)
```

List Elements: [1, 2, 3, 4, 5]

Sum of Elements: 15

```
[5]: def multiplier(n):  
      return lambda x: x * n  
double = multiplier(2)  
triple = multiplier(3)  
print("Double of 5:", double(5))  
print("Triple of 5:", triple(5))
```

Double of 5: 10

Triple of 5: 15

```
[6]: students = [  
      ("Anu", 85),  
      ("Rahul", 72),  
      ("Priya", 90),  
      ("Kiran", 78)  
]  
sorted_students = sorted(students, key=lambda x: x[1])  
print("Students Sorted by Marks:")  
for student in sorted_students:  
    print(student)
```

Students Sorted by Marks:

('Rahul', 72)

('Kiran', 78)

('Anu', 85)

('Priya', 90)

```
*[7]: names = ["anu", "rahul", "priya", "kiran"]
uppercase_names = list(map(lambda x: x.upper(), names))
print("Original Names:", names)
print("Uppercase Names:", uppercase_names)
```

Original Names: ['anu', 'rahul', 'priya', 'kiran']

Uppercase Names: ['ANU', 'RAHUL', 'PRIYA', 'KIRAN']

```
[8]: numbers = [-10, 20, -30, 40, 50, -60]
positive_numbers = list(filter(lambda x: x > 0, numbers))
print("Original List:", numbers)
print("Positive Numbers:", positive_numbers)
```

Original List: [-10, 20, -30, 40, 50, -60]

Positive Numbers: [20, 40, 50]

```
[9]: from functools import reduce
numbers = [12, 45, 67, 23, 89, 34]
```



```
print("Sum:", result)
```

```
First Number: 10
Second Number: 20
Sum: 30
```

```
[11]: words = ["Python", "Java", "C", "JavaScript"]
      lengths = list(map(lambda x: len(x), words))
      print("Words:", words)
      print("Lengths:", lengths)
```

```
Words: ['Python', 'Java', 'C', 'JavaScript']
Lengths: [6, 4, 1, 10]
```

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[ ]:
```